

Regional Sales Performance Analysis

About the Company- Superstore is a Canadian e-commerce retailer selling consumer goods to individual and corporate customers across 8 regions. Between 2009 and 2012, Superstore served approximately 1,460 unique customers and generated roughly \$14.9M in total sales.

1. Client Background

Superstore is a Canadian e-commerce retailer selling consumer goods across 8 regions, from densely populated provinces like Ontario and Quebec to sparsely populated northern territories like Nunavut and Yukon.

Superstore's book of business spans four years of transactions (2009–2012) across pricing, discounting, and customer activity by region. As a Data Analyst supporting the regional pricing team, my responsibility is to analyze this data and translate the findings into a clear, actionable recommendation for the Regional Pricing Manager.

The Analysis focuses on two business question:

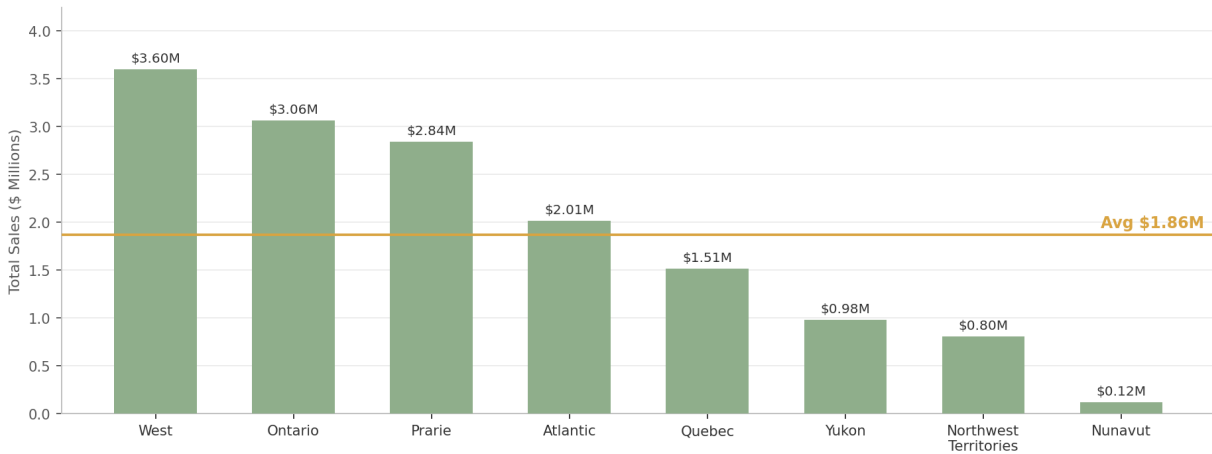
- **Pricing:** Should we lower prices further in regions with low total sales like Nunavut to try to increase demand?
- **Investment:** Of all 8 regions which markets should we prioritize for additional marketing and sales investment.

North Star Metrics

- **Average Unit Price:** Tracked by region and year to test if price changes actively drive customer demand.
- **Total Sales & Unique Customers:** Measured to understand the actual market size and revenue contribution of each region.
- **Customers per 100,000 Population:** Calculated as customers per 100,000 population by enriching sales data with official census data. This was the ultimate metric that shifted the entire business strategy.

2. Executive Summary

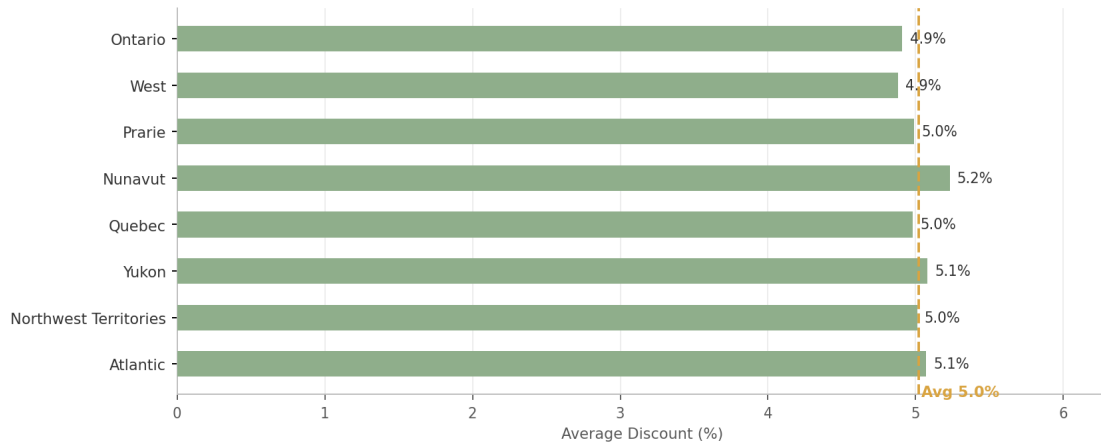
Total Sales by Region (2009-2012)



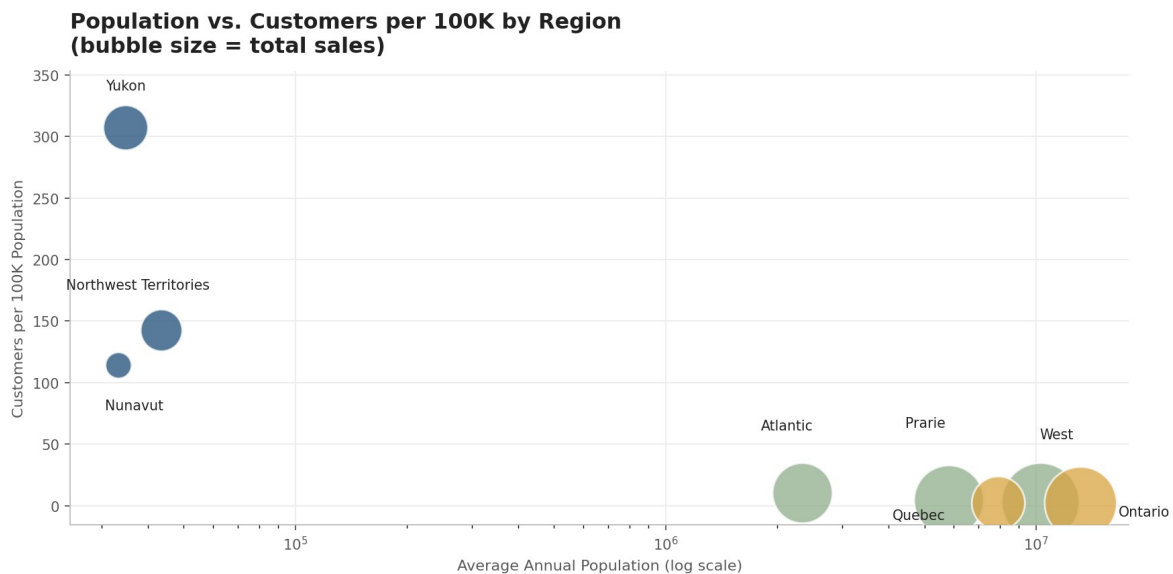
Key Insights & Findings

- **Pricing Hypothesis Disproved:** Nunavut's average unit price dropped by 75% (2009-2012) but sales only grew by 34% proving that lowering prices does not drive demand in this region. Average discount rates were also similar across all 8 regions (4.7%-5.7%),

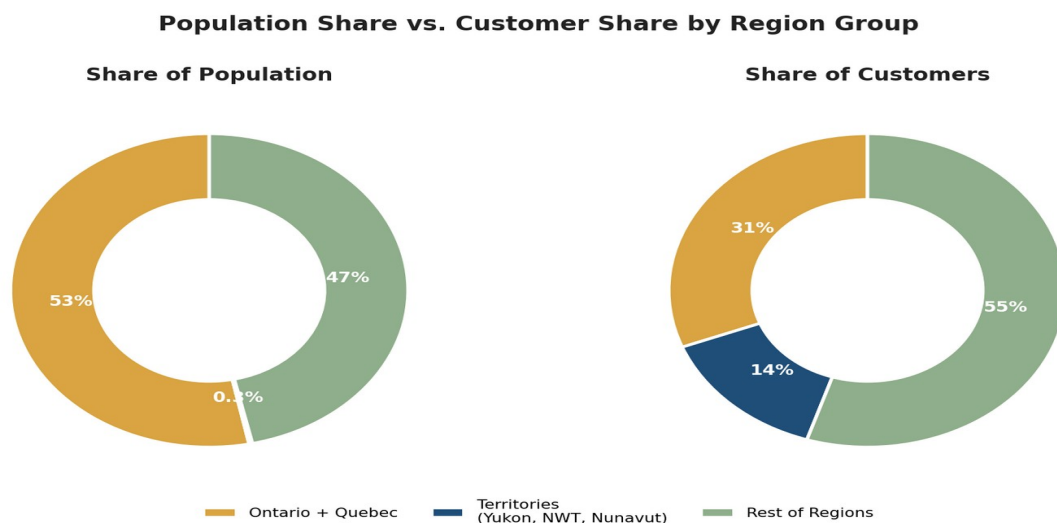
Average Discount Rate by Region — Nearly Identical Everywhere



- **Customer Count Shows a Strong Relationship with Sales:** Northwest Territories consistently outsold Nunavut by 5x-10x driven by having 3x more unique customers not lower prices.
- **Nunavut Has a Relatively High Customer-to-Population Ratio:** At 114 customers per 100K population, Nunavut ranks 3rd highest despite having the lowest sales — this shows that its low sales are related to the size of its market rather than low customer reach or high prices.



- **Ontario & Quebec show the Largest Growth Opportunities:** Both provinces have the lowest customers per 100K (around 2, vs. 100+ in the territories) despite having Canada's largest populations



- **Yukon Has the Highest Customer-to-Population Ratio:** Yukon achieved the highest rate nationwide (307 customers per 100K).

3. Dataset Structure

The analysis draws on the Global Superstore transactional dataset (2009–2012), queried directly in SQL Server, and covers Canadian sales across 8 regions with roughly 8,400 order records. Key fields used in this analysis:

- Sales metrics: unit price, discount, sales revenue, order quantity
- Customer information: customer name, region, order date
- Population data: official population estimates for each year of the 2009–2012 sales period (from Statistics Canada census data), averaged across the four years, joined to the sales data separately to calculate customers per 100,000 population. This data is not part of the original sales dataset.

```

1  select s.region,
2  count(distinct s.customer_name) as unique_customers,
3  p.population
4  from SuperStore s
5  join region_population p on s.region=p.region
6  group by s.region, p.population
7  order by p.population desc;

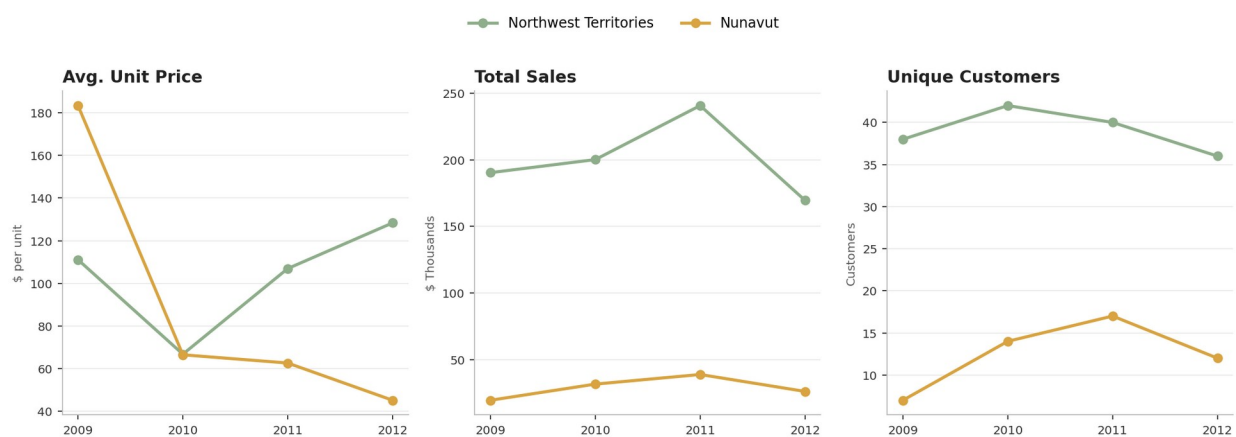
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- Output: the joined data was used to calculate customers per 100,000 population for each region and compare market reach across regions.

4. Deep Insights

Testing Decision 1: Does Price Explain the Sales Gap?

Nunavut and Northwest Territories were compared directly, as both are small markets with similar populations but very different sales outcomes.

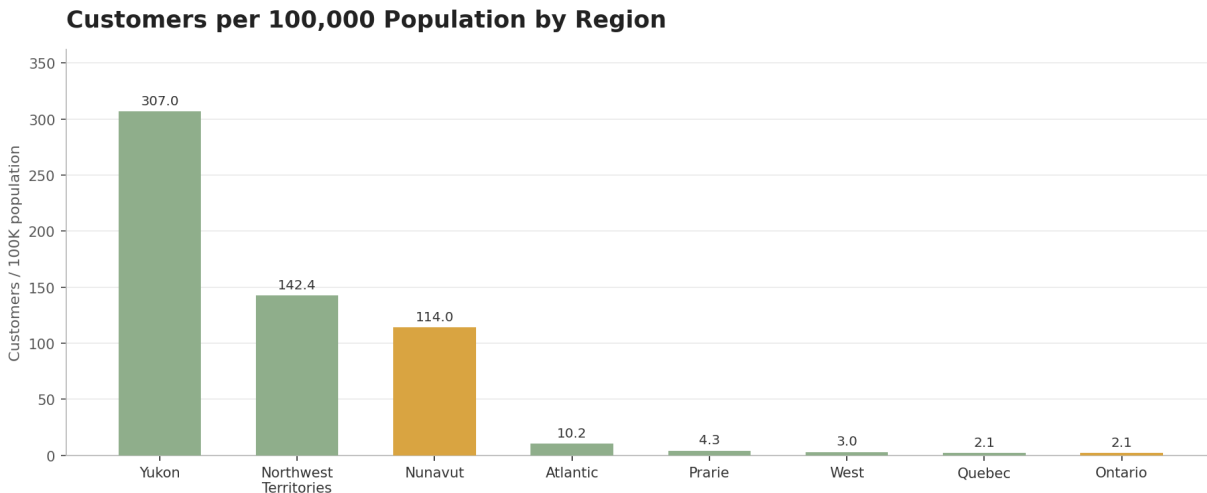


- Price moved in the opposite direction of what was predicted: Nunavut's price fell (2009–2012) while Northwest Territories' price went up yet Northwest Territories consistently outsold Nunavut by 5-10x every single year.
- Unique customers count shows a stronger relationship with total sales than price changes do.

- Price does not appear to be the primary driver of the sales gap between these two regions; the number of customers being reached is.

Testing Decision 2: Identifying Regional Growth Opportunities

To understand why customer counts differ so much between regions official population data was joined to the sales data to calculate customers per 100,000 population for every region not just Nunavut and Northwest Territories.



- **Nunavut is not underperforming — it has a relatively high number of customers per capita.** At 114 customers per 100K population, Nunavut ranks 3 out of 8 regions despite having the lowest total sales in absolute terms.
- **Ontario and Quebec have the largest potential.** Both rank lowest in customers per 100K (around 2, vs. 100+ in the territories) despite having the largest populations in Canada meaning most of their population is not reflected in the customer base.
- **Yukon is the benchmark to study.** At 307 customers per 100K it has the highest rate in the country and may hold a repeatable playbook for the other territories.

5. Recommendations

With price ruled out as the primary driver (Decision 1), the recommendations below answer Decision 2: where to focus investment.

Priority	Recommendations	Evidence	Why?	Expected Impact
P0	Do not lower prices further in Nunavut.	Customers per 100K: 114 customers per 100K population – 3rd highest in Canada.	The region already has a relatively high number of customers per capita compared to its small population; any further price reduction may reduce margins.	Protects profit margin in an already-saturated small market.
P0	Redirect marketing investment toward Ontario	Customers per 100K: 2.13	Both regions combine a very large population with	These regions appear to have the

Priority	Recommendations	Evidence	Why?	Expected Impact
	and Quebec.	(Ontario) and 2.15 (Quebec) customers per 100K population – the lowest in Canada despite the largest populations.	very few customers per capita.	largest opportunity for increasing customer reach.
P1	Study the Yukon-go-to market approach and evaluate replicating it elsewhere.	Customers per 100K: 307 customers per 100K population – the highest in Canada.	Understanding what is driving Yukon's outsized customer-to-population ratio (channel mix, local partnerships, marketing approach) could inform strategy for Ontario/Quebec expansion.	The findings could help identify ways that we could apply to other underpenetrated regions.

6. Limitations

- Population figures are averaged across 2009–2012 and mapped to the dataset's aggregated region labels (e.g., "Atlantic," "Prairie," "West"); the exact provinces included in each grouping were inferred and would benefit from confirmation against Superstore's own regional definitions.
- The dataset covers 2009–2012 only; more recent data would be needed to confirm these patterns still hold today.
- Product-level margin data (Product_Base_Margin) was excluded from this analysis, as profitability by product was outside the scope of the pricing-versus-demand question being investigated.

7. Next Steps

- Study the factors behind Yukon's high customer reach to identify practices or market characteristics that may provide useful insights for other regions.
- Explore customer reach opportunities in Ontario and Quebec, which have large populations but relatively low customer reach.
- Extend the analysis to the product-category level to identify which categories contribute most to the customer gap in Ontario and Quebec.